

QUESTION BANK

UNIT-1

1. Explain the contribution of Aryabhata to Indian mathematics.
2. What are the four parts of Aryabhatiya? Briefly describe each.
3. What is the importance of Brahmagupta's work *Brāhmasphuṭasiddhānta*?
4. How did Bhaskaracharya's work in *Lilavati* and *Bijaganita* influence mathematics?
5. Explain the mathematical achievements of Srinivasa Ramanujan in number theory.
6. Discuss the role of Swami Bharati Krishna Tirthaji Maharaj in rediscovering Vedic Mathematics.
7. Explain the sutra *Ekadhikena Purvena* with two step-by-step examples.
8. Describe the use of *Nikhilam Navatashcaramam Dashatah* in subtraction with an example.
9. State the meaning and application of the sutra *Urdhva Tiryagbhyam* in multiplication. Give an example.
10. How does the sutra *Paravartya Yojayet* simplify division? Illustrate with a base 100 example.
11. Explain the application of *Shunyam Samyasamuccaye* in solving algebraic expressions. Provide one example.
12. What is the condition for applying the sutra *Anurupye Shunyamanyat* in linear equations? Give a suitable example.
13. Describe how the sutra *Ekanyunena Purvena* is used for multiplying numbers like 99×97 .

UNIT-2

1. Successive Subtraction from Left with Borrowing Adjustment

1) $742 - 289 = ?$

2) $563 - 378 = ?$

3) $894 - 496 = ?$

4) $678 - 459 = ?$

5) $932 - 587 = ?$

2. Find squares of following numbers by using "Ekādhikena Pūrvena" (एकाधिकेन पूर्वेण):

1) 25

2) 35

3) 45

4) 55

5) 105

6) 85

7) 95

8) 75

9) 65

3. find decimal expression of following numbers.

1) $\frac{1}{19}$

2) $\frac{1}{29}$

3) $\frac{1}{7}$

4) $\frac{1}{13}$

5) $\frac{1}{81}$

4. Find Complement of 103 with respect to 100.

5. Find complement of 86 with respect to 100.

6. Find complement of 467 with respect to 1000.

7. Find Complement of 1005 with respect to 10000.

8. Multiply 98×97 by using complement.

9. Multiply 102×104 by using complement.

10. Find the Digital Root of 986.

11. Check a multiplication using Digital Roots if $47 \times 86 = 4042$ is correct?
12. Check a division using Digital Roots if $828 \div 9 = 92$ is correct?
13. Find the digital root of $(256 \times 18) + 37$.
14. Construct the table of **9** using the Nikhilam Sutra.
15. Derive the table of **11** using the "digit middle insert" technique.
16. Show how Urdhva-Tiryagbhyam helps find 84×13 .
17. Create a 10-step multiplication table of **99** using Runak method.
18. Construct a 10-step multiplication table of **98** using Runak logic.
19. Find the value of 999×5 using complement method.
20. Find 9999×7 using Runak method.
21. Describe divisibility by 3, 7, 11, 13, 17, 19 by 3 examples.
22. Check if 1012 is divisible by 99.
23. Check if 1001 is divisible by 101.
24. Is 2997 divisible by 999?
25. Check if 532 is divisible by 7 using Osculation Method.
- 26.** Check if 858 is divisible by 13 using Osculation Method..
27. Check if 874 is divisible by 19 using Osculation Method..

Unit-3

❖ Using निखिलं नवतश्चरमं दशतः (Nikhilam Navataścaramam Daśatah)

- 1) 9×8
- 2) 7×9
- 3) 11×12
- 4) 13×9
- 5) 98×97
- 6) 94×96
- 7) 101×102
- 8) 103×99
- 9) 107×104
- 10) 92×95
- 11) 998×997
- 12) 1004×1002
- 13) 996×1003
- 14) 1007×999
- 15) 198×197
- 16) 203×202
- 17) 48×47
- 18) 52×51
- 19) 298×297
- 20) 304×301

❖ Using Ekanyunena Purvena (एकान्यूननेन पूर्वेण)

- 1) 999×738
- 2) 999×999

- 3) 9999×4567
- 4) 9999×9999
- 5) 99999×63048
- 6) 999×874
- 7) 999×507
- 8) 9999×8321
- 9) 999×901
- 10) 9999×2034
- 11) 99999×99999
- 12) 999×667
- 13) 99999×54321
- 14) 999×212
- 15) 9999×4873
- 16) 9999×7643
- 17) 999×893
- 18) 99999×10000
- 19) 9999×1000
- 20) 999×900

❖ Multiplication with अन्त्ययोर्दसाकेनप

- 1) 42×48
- 2) 63×67
- 3) 51×59
- 4) 94×96
- 5) 71×79
- 6) 25×25
- 7) 84×86
- 8) 103×107
- 9) 812×818

- 10) 1003×1007
- 11) 47×43
- 12) 64×66
- 13) 72×78
- 14) 38×32
- 15) 115×115
- 16) 99×91
- 17) 54×56
- 18) 73×77
- 19) 106×104
- 20) 10006×10004

❖ Urdhva-Tiryagbhyam (ऊर्ध्व-तिर्यग्भ्याम्)

- 1) 123×456
- 2) 314×271
- 3) 738×642
- 4) 981×327
- 5) 856×749
- 6) 675×843
- 7) 432×219
- 8) 593×874
- 9) 386×729
- 10) 917×583
- 11) 23×14
- 12) 41×32
- 13) 67×48
- 14) 56×39
- 15) 89×76
- 16) 34×21
- 17) 72×53
- 18) 91×45
- 19) 63×27
- 20) 88×44

❖ Ekadhikena Purvena (एकाधिकेन पूर्वेण)

- 1) $(15)^2$
- 2) $(35)^2$
- 3) $(45)^2$
- 4) $(55)^2$
- 5) $(75)^2$
- 6) $(85)^2$

- 7) $(95)^2$
- 8) $(125)^2$
- 9) $(205)^2$
- 10) $(305)^2$

❖ **यावद्गुणम् (Yāvadūnam)** method of squaring numbers.

- 1) 98^2
- 2) 97^2
- 3) 94^2
- 4) 89^2
- 5) 995^2
- 6) 9998^2
- 7) 78^2
- 8) 88^2
- 9) 96^2
- 10) 992^2
- 11) 102^2
- 12) 105^2
- 13) 112^2
- 14) 109^2
- 15) 1005^2
- 16) 1008^2

❖ **द्वन्द्वयोग (Dwandwa Yoga)**

➤ **2-digit numbers**

- 1) 34^2
- 2) 58^2
- 3) 76^2
- 4) 89^2
- 5) 92^2

➤ **3-digit numbers**

- 1) 123^2
- 2) 247^2

- 3) 365^2
- 4) 489^2
- 5) 512^2

➤ **4-digit numbers**

- 1) 1024^2
- 2) 2135^2
- 3) 3247^2
- 4) 4506^2
- 5) 5678^2

➤ **Mixed challenge (random difficulty)**

- 1) 703^2
- 2) 906^2
- 3) 1789^2
- 4) 2351^2
- 5) 8999^2

❖ **Square root of a perfect square**

- 1) 20736
- 2) 106929
- 3) 53361
- 4) 324900
- 5) 15876
- 6) 79524
- 7) 4489
- 8) 529
- 9) 784
- 10) 5929
- 11) 1444
- 12) 1210000
- 13) 8836
- 14) 275625
- 15) 15129
- 16) 16900
- 17) 207025
- 18) 17424
- 19) 52900
- 20) 3481

UNIT-4

1. Find addition of polynomials $3x^2 + 2x + 4$ and $2x^2 + 3x + 2$.
2. Find addition of polynomial $4y^2 + 3y + 1$ and $-2y^2 - y + 4$.
3. Find addition of polynomial $5x^5 + 6x^3 + 8x + 100$ and $10x^2 + 11x + 11$.
4. Find addition of polynomial $x^2 + x + 1$ and $x^3 - x - 2$.
5. Subtract $4x^3 + 3x^2 + 2x + 3$ from $5x^3 + 4x^2 + 7$.
6. Subtract $-3x^2 + 50x - 2$ from $4x^3 - x^2 + 7$.
7. Subtract $x - 5$ from $4x^4 - 5x^3 + 6x^2 + 8$.
8. Subtract x from $x^2 + 1$.
9. Subtract $x^2 - 2$ from $x^4 + x^2 + 1$.
10. Use the Urdhva Triyabhyam sutra to multiply $2x + 1$ and $3x + 3$.
11. Use the Urdhva Triyabhyam sutra to multiply $4x + 2$ and $2x + 5$.
12. Use the Urdhva Triyabhyam sutra to multiply $2x + 1$ and $3x + 3$.
13. Use the Urdhva Triyabhyam sutra to multiply $4x^2 + 2x + 3$ and $3x^2 + 3x + 2$.
14. Use the Urdhva Triyabhyam sutra to multiply $7x^2 + 8x + 2$ and $8x^2 + 6x + 5$.
15. Use the Urdhva Triyabhyam sutra to multiply $2x^2 + 3x + 4$ and $5x^2 + 6x + 7$.
16. Use the Urdhva Triyabhyam sutra to multiply $x^2 + 5$ and $x^2 - 11x + 6$.
17. Use the Urdhva Triyabhyam sutra to multiply $2x - 5$ and $x - 1$.
18. Use the Urdhva Triyabhyam sutra to find square of $2x - 1$.
19. Use the Urdhva Triyabhyam sutra to find square of $5x^2 - 2x - 1$.
20. Use the Urdhva Triyabhyam sutra to multiply $(3x - 2)$, $(x - 1)$ and $(3x + 5)$.
21. Use the Urdhva Triyabhyam sutra to find cube of $2x + 1$.
22. Use the Urdhva Triyabhyam sutra to find cube of $x - 2$.
23. Use the Urdhva Triyabhyam sutra to multiply $(2x + 1)$, $(3x + 3)$ and $(x - 5)$.
24. Use the concept of Paravartya to find $\frac{2x^2+4x+7}{x+2}$.
25. Use the concept of Paravartya to find remainder and quotient of $\frac{3x^3+2x^2+5x+9}{x-1}$.
26. Use the concept of Paravartya to find remainder and quotient of $\frac{4x^2-4}{x-1}$.
27. Use the concept of Paravartya to find remainder and quotient of $\frac{x^2+1}{x-1}$.
28. Use the concept of Paravartya to find remainder and quotient of $\frac{x^5+x^3+x-1}{x-5}$.

29. Use the concept of Paravartya to find remainder and quotient of $\frac{x^3-1}{x+1}$.
30. Use the concept of Paravartya to find remainder and quotient of $\frac{x^2-2x+1}{x-1}$.
31. Factorize quartile $4x^2 + 12x + 5$ use the concept of Anurupyena and
Adyamayenantyamanteya.
32. Factorize quartile $15x^2 - 14xy - 8y^2$ use the concept of Anurupyena and
Adyamayenantyamanteya.
33. Factorize quartile $3x^2 + 14x - 5$ use the concept of Anurupyena and
Adyamayenantyamanteya.
34. Factorize quartile $6x^2 - 23x + 7$ use the concept of Anurupyena and
Adyamayenantyamanteya.
35. Factorize quartile $8x^2 - 22x + 5$ use the concept of Anurupyena and
Adyamayenantyamanteya.
36. Factorize quartile $12x^2 - 23xy + 10y^2$ use the concept of Anurupyena and
Adyamayenantyamanteya.
37. Factorize polynomial $3x^2 + 7xy + 2y^2 + 11xz + 7yz + 6z^2$ use the concept
of Lopana Sthapnabyam.
38. Factorize polynomial $12x^2 + 11xy + 2y^2 - 13xz - 7yz + 3z^2$ use the
concept of Lopana Sthapnabyam.

39. Factorize polynomial $3x^2 + 6y^2 + 6z^2 + 11xy + 7yz + 6xz + 19x + 22y + 13z + 20$ use the concept of Lopana Sthapnabyam.
40. Factorize the polynomial $2x^3 - 3x^2 + 1$.
41. Factorize the polynomial $x^3 - 3x^2 - 9x - 5$.

UNIT-5

1. Find HCF(12,15,21) and LCM(12,15,21).
2. Find HCF(50,70,85) and LCM(50,70,85).
3. Find HCF(10,30,50,100) and LCM(10,30,50,100).
4. Find HCF(8,20,32) and LCM(8,20,32).
5. Find HCF(10,15,100) and LCM(10,15,100).
6. Find HCF(3,9,18,27) and LCM(3,9,18,27).
7. Find LCM(10,20,30) and HCF(10,20,30).
8. Find LCM(5,10,15) and HCF(5,10,15).
9. Find LCM(11,12,13) and HCF(11,12,13).
10. Find LCM(10,13,20) and HCF(10,13,20).
11. Find LCM(6,12,18) and HCF(6,12,18).
12. Solve following bivariate linear equations.
 - 1) $3x + 5y = 19$
 - 2) $2x + 3y = 12$
13. Solve following bivariate linear equations.

- 1) $3x - 5y = 19$
- 2) $-7x + 3y = -1$

14. Solve following bivariate linear equations.

- 1) $x + y = 8$
- 2) $2x - 3y = 1$

15. Solve following bivariate linear equations.

- 1) $11x + 15y = -23$
- 2) $7x - 2y = 20$

16. Solve following bivariate linear equations.

- 1) $2x + y = 5$
- 2) $3x - 4y = 2$

17. Solve following bivariate linear equations.

- 1) $5x - 3y = 11$
- 2) $6x - 5y = 9$

18. Solve following bivariate linear equations.

- 1) $11x + 6y = 28$
- 2) $7x - 4y = 10$

19. Solve following bivariate linear equations.

- 1) $6x + 7y = 8$
- 2) $19x + 14y = 16$

20. Solve following bivariate linear equations.

- 1) $12x + 8y = 7$
- 2) $16x + 16y = 14$

21. Solve following bivariate linear equations.

- 1) $12x + 78y = 12$
- 2) $16x + 96y = 16$

22. Solve equation $\frac{3x+4}{6x+7} = \frac{5x+6}{2x+3}$.

23. Solve equation $\frac{1}{x+7} + \frac{1}{x+9} = \frac{1}{x+6} + \frac{1}{x+10}$.

24. Solve equation $\frac{1}{x-7} + \frac{1}{x+9} = \frac{1}{x+11} + \frac{1}{x-9}$.

25. Solve equation $\frac{1}{x-8} + \frac{1}{x-9} = \frac{1}{x-5} + \frac{1}{x-12}$.

26. Solve equation $\frac{x-2}{x-3} + \frac{x-3}{x-4} = \frac{x-1}{x-2} + \frac{x-4}{x-5}$.

27. Solve equation $\frac{x}{x-2} + \frac{x-9}{x-7} = \frac{x+1}{x-1} + \frac{x-8}{x-6}$.

28. Solve equation $\frac{2x-3}{3x-8} + \frac{3x-20}{4x-35} = \frac{x-3}{2x-9} + \frac{4x-19}{5x-34}$.

29. Solve equation $\frac{x-2}{3x-8} + \frac{x-7}{4x-35} = \frac{x-4}{2x-9} + \frac{x-5}{5x-34}$.

30. Solve equation $\frac{x-3}{3x-13} + \frac{x-9}{4x-41} = \frac{x-5}{2x-13} + \frac{x-7}{5x-41}$.

31. Solve the equation $\frac{2}{2x+3} + \frac{3}{3x+2} = \frac{1}{x+1} + \frac{6}{6x+7}$.

32. Solve equation $\frac{3}{3x+1} - \frac{6}{6x+1} = \frac{3}{3x+2} - \frac{2}{2x+1}$.

33. Solve equation $\frac{3}{3x+1} + \frac{2}{2x-1} = \frac{3}{3x-2} + \frac{2}{2x+1}$.